Global Risk Consultants®

Global Risk Consultants Corp.
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May 27, 2024

Najem Almamo
Planner & Scheduler for Preventive Maintenance
Cargill Incorporated
Starches and Sweeteners
Research & Development
84 Havenstraat
Vilvoorde, Belgium 1800

Re: Thermographic Survey - May 27, 2024
Research & Development - Vilvoorde, Belgium
File No. 495.1518
Client No. 10779

Dear Najem Almamo:

This report documents the results of my recent survey of the above-referenced location. Thermal imaging, ultrasonic listening and other technologies were utilized during this survey to locate abnormal electrical and mechanical conditions at your facility, which were indicative of impending failure. This report contains a list of equipment surveyed, documentation of detected recommendations and a summary of those recommendations.

Note that TÜV SÜD Global Risk Consultants augments our infrared thermographic inspection by utilizing state-of-the-art ultrasonic (airborne/structure-borne) listening equipment during the survey of your facility's and utility's electrical system. Our ultrasound technology allows us to listen for thermally undetectable corona and tracking problems in your facility's medium- or high-voltage equipment. It also allows us to listen for arcing in high-voltage switchgear that could not be opened for infrared inspection. Further, it is strategically utilized in low-voltage switchgear and/or transformers for anomalies mostly pertaining to mechanical looseness or contaminants.

As a result of this study of your plant's electrical power distribution and selected mechanical equipment, I report the detection of two (2) recommendations. The specifics of these recommendations are contained in the attached Recommendation Summary Report and detailed on an individual recommendation report page.

Also, please note that both of the recommendations were appointed an FYI - "For Your Information" designation. These visually detected recommendations serve to bring to your attention conditions observed during the study, which may have the potential for negative impact on the integrity of the plant's operating systems. They document previously unknown conditions of risk for reference and to assist in the mitigation of problem.

Of the two (2) recommendations detected during our last inspection, one (1) was found to still exhibit abnormal condition. As discussed, Recommendation No. 2023-04-2 of this report warrants special attention during repair and/or possible replacement.

As discussed during our closeout meeting:

1. An "Impact" rating is assigned to each detected recommendation. It is derived from our on-site discussions regarding the potential ramification each recommendation would have on business interruption and potential asset loss associated with the recommendations, should it reach a failure mode. To be specific, this is not a rating based on a recommendation's temperature. It is a rating that reflects the importance of the recommendation's potential damage to the plant's ongoing production and asset security. Impact ratings for detected faults are assigned after consultation with pertinent plant personnel during the on-site closeout meetings.

This report does not purport to set forth all hazards nor to indicate that other hazards do not exist. However, we urge that recommendations presented, if any, be given serious consideration. We would be pleased to discuss alternative solutions.

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2. Both of the recommendations contained within this report presently fall within the realm of normal maintenance issues; for example: hot connections, defective components, deteriorated equipment, etc. As discussed, they require no special tools or repair technique to correct at this time. Each recommendation has been assigned a severity coding to assist you in the prioritization of repair.

3. As discussed in our closeout meeting, Recommendation Nos. 2024-05-1 depict conditions that have the present potential for considerable asset loss and business interruption. These conditions have been assigned a high impact rating and warrant your immediate attention.

Please note that your Corporate Risk Management Department requests that you review the recommendations and respond to each within 30 days of receiving your final report. GRC will send an email containing an electronic link within one to two business days of receiving your final report, which provides for an online response for each recommendation. Full instructions are also provided in the email. If there are no recommendations for this survey, you will not be receiving the response email.

Please convey my thanks to all your staff involved with this survey for the co-operation and assistance I received during this inspection. Should you have any questions regarding this report or should you require secondary a follow-up or emergency inspection; please contact me at yannick.ocke@tuvsud.com.

Sincerely,

Yannick Ocke

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THERMOGRAPHIC SURVEY

Prepared for

Cargill Incorporated

Starches and Sweeteners
Research & Development
Vilvoorde, Belgium

File No. 495.1518
Client No. 10779

Accompanied By: Danny, Electrician

Exit Interview With: Najem Almamo, Planner & Scheduler for Preventive Maintenance

Thermographer: Yannick Ocke

May 27, 2024

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Equipment Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 390309
Date Completed: 27-May-24

Seq. No.	Status	Equipment	Location	Equip. ID
1	Inspected	HV Switches Cell 1 (SF-6, US Only)	Incoming Substation	121
2	Inspected	HV Switches Cell 2 (SF-6, US Only)	Incoming Substation	173
3	Inspected	HV Switches Cell 3 (SF-6, US Only)	Incoming Substation	174
4	Inspected	HV Switches Cell 4 (SF-6, US Only)	Incoming Substation	175
5	Inspected	HV Switches Cell 5 (SF-6, US Only)	Incoming Substation	122
6	Inspected	HVP 1-1 To 1-4	Incoming Substation	123
7	**Fault Info**	Electrical	General	171

Find No.: 2 — **Equipment:** Many Cable Penetrations
Rec. No.: 2024-05-1 — **Specific Item Detail:** Abnormal visual condition re: lack of fire stop packing.
Severity: FYI — **Probable Cause:** Potential for collateral fire damage.
History: New — **Recommendation:** Fireproof any unprotected openings with UL Listed fireproofing media having a minimum resistance rating of 1-hour
Impact: High — **Redundancy Spare:** No Spare

Find No.: 1 — **Equipment:** Panel Filters
Rec. No.: 2023-04-2 — **Specific Item Detail:** Abnormal visual condition re: dust/dirt contamination.
Severity: FYI — **Probable Cause:** Dirty filters blocking the airflow.
History: PFR — **Recommendation:** Filters should be cleaned or replaced frequently to avoid high temperatures in the panels.
Impact: Low — **Redundancy Spare:** Spare - Inline Redundancy

8	Removed	LV Switch X2 400A	B-1	10
9	Removed	VB.B01 Panel	B-1	11
10	Removed	A414-2 Panel	B-1	82
11	Removed	B430-4 Panel	B-1	83
12	Removed	VB.B11 Panel	B-1	12
13	Removed	VB.B21 Panel	B-1	13
14	Removed	Panel	B-1	14
15	Removed	Technisch Verdiep Panel	B-1	15
16	Removed	Computer Panel	B-1	16
17	Removed	Lift "B" Panel	B-1	17
18	Removed	Vb. CO1 Panel	B-1	18
19	Removed	Fermantatie Panel	B-1	19
20	Removed	Bord O.P.P	B-1	20
21	Removed	Bord B100-11A	B-1	21

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Seq. No.	Status	Equipment	Location	Equip. ID
22	Removed	Bord B100-11B	B-1	22
23	Inspected	B100-12 Distribution Board	B-1, Basement	84
24	Inspected	Fire Evacuation Alarm System C.P.	B-1, Basement	152
25	Inspected	Fire Evacuation Alarm System Batteries	B-1, Basement	153
26	Inspected	VBB.01 Switch Panel (2)	B050	9
27	Inspected	LV-B0-51-001-22	B0	178
28	Inspected	LV-B0-51-002-12	B0	179
29	Inspected	B111-1 Bord CTHR	B111	23
30	Inspected	B136-1 Computer Supply	B136, Server Room	24
31	Inspected	Servers	B136, Server Room	172
32	Inspected	UPS1	B136, Server Room	176
33	Inspected	UPS2	B136, Server Room	177
34	Inspected	VBB.1.1 Distribution Panel	B150	25
35	Inspected	VBB.2.1 Distribution Panel	B250	26
36	Inspected	VBB.3.1 Distribution Panel	B350	27
37	Inspected	C335-1 Bord Stookplaats	C335	37
38	Inspected	B430-1 HVAC Control Panel, LV_B+4-30-001-13	B430	28

Find No.: 501 **Equipment:** 39Q13
Rec. No.: 2023-04-1 **Specific Item Detail:** Abnormal thermal condition re: the breaker.
Severity: *** **Probable Cause:** Overloading suspected.
History: PFC **Recommendation:** Check load versus rated capacity. Ensure proper cooling.
Impact: Low **Redundancy Spare:** Spare - Inline Redundancy

39	Inspected	B430-2 AC Supply Panel	B430	85
40	Inspected	Fume Extraction Fans (~40)	B430	151
41	Inspected	VBC.0.1 Supply Panel	C030 Confectionery	30
42	NIS/NOP	Busbar	C030 Confectionery	148
43	Inspected	VBC.2.1/C103-1 Supply Panel	C130	31
44	Inspected	C130-2 Bord Collector	C130	32
45	Inspected	C130-4 Bord PG5EG5	C130	33

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Facility: Research & Development
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File No.: 495.1518
Client No.: 10779
Report ID: 390309
Date Completed: 27-May-24

Seq. No.	Status	Equipment	Location	Equip. ID
46	Inspected	Bord C130-3 PG4-EG4	C130	34
47	Inspected	C130-5 Freezer Control Panel	C130	35
48	Inspected	Freezer Door	C130	149
49	Inspected	VBC.3.1 C230-1SUPPLY Panel	C230	36
50	Inspected	Busbar	C230	150
51	Inspected	A008-1 Main Supply	A008 Toilets	1
52	Inspected	A001-1 Distribution Panel	A001, Reception	88
53	Inspected	Bord A108-1	A108	2
54	Inspected	Bord A208-1	A208	3
55	Inspected	Bord A308-1	A308	4
56	Inspected	A414-1 HVAC Control Panel	A414	7
57	Inspected	A414-3 Distribution Panel	A414	86
58	Inspected	A414-2 Distribution Panel	A414	87
59	Inspected	A404-1 Liftkamer	A404	5
60	Removed	R209-1 Distribution Panel	R209	90
61	Removed	R003-1	R003	89
62	Inspected	Verdeelbord Meat	D Pilot Meat and Pharma	168
63	Inspected	Verdeelbord Pharma	D Pilot Meat and Pharma	169
64	Inspected	Verdeelbord Meat 2	D Pilot Meat and Pharma	170
65	Inspected	J050-1 Distribution Panel (1-2)	J050	40
66	Inspected	J050-1-1 Distribution Panel (2)	J050	180
67	Inspected	Busbar 250	J050	147
68	NIS/NOP	Compressor I0021	J033	145
69	NIS/NOP	Bord J1.30	J1	181
70	Inspected	Bord LV_J+1_30_U549_001_13	J1	184
71	Inspected	Bord LV_J+2_30_U547_001_13	J2	185
72	Inspected	Busbar	J2	182
73	NIS/NOP	Busbar	J304	42

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Facility: Research & Development
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File No.: 495.1518
Client No.: 10779
Report ID: 390309
Date Completed: 27-May-24

Seq. No.	Status	Equipment	Location	Equip. ID
74	Inspected	K.0.80-01, Lighting Control Panel	K0, Under Stairs	100
75	Inspected	K.1.19.01, HVAC	K1 Technical Room	101
76	Inspected	TC AIR-SB02	K1 Technical Room	102
77	Inspected	Weiss Panel	K1 Technical Room	103
78	Inspected	Compressor Koelkamer 1-2	K1 Technical Room	143
79	Inspected	Compressor Diepvries	K1 Technical Room	144
80	Inspected	K.2.28-01, Office	K2.10	104
81	Inspected	K.3.01-01, Labo	K3.12	105
82	Inspected	Bord LV_J+3.36_001_13	J3	183
83	Inspected	Bms/Noodverlichting	P3, K3.07 Fermentation	106
84	Inspected	Panel Hosokawa Dryer	P3	166
85	Inspected	Teblick Panels	P3	107
86	Inspected	AHU Panels (Panels 1-2)	P3	108
87	Inspected	AHU Control Panel Small	P3	141
88	NIS/NOP	Oil Heater 10803	P3	142
89	Inspected	TD HVAC	P3	167
90	Inspected	P.2.50-1 Panels 1-3	P3	109
91	Inspected	P.2.50-2 Panels 1-2	P3	110
92	Inspected	Voedingsbord 140203HX003	P3	139
93	Inspected	P3.50.01	P3	140
94	Inspected	140204-2	P2 Technical Corridor	111
95	Inspected	140203-1	P2 Technical Corridor	134
96	Inspected	140204-1	P2 Technical Corridor	112
97	NIS/NOP	CM20 142201-1	P2 Technical Corridor	135
98	Inspected	Panel P+2_50 U2257	P2 Technical Corridor	136
99	Removed	Extruder U0049 MCC1	P2 Technical Corridor	137
100	Inspected	CB20 141201-1	P2 Technical Corridor	113
101	Inspected	Door Control	P2 Corridor, P2.92	138

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Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
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Seq. No.	Status	Equipment	Location	Equip. ID
102	Inspected	Busbar MDP1/29Q1	P2.20	133
103	Not Inspecte	Compressor 1.0701 (Interlocked)	P3, Compressor Room	99
104	Not Inspecte	Compressor 1.0702 (Interlocked)	P3, Compressor Room	129
105	Inspected	Filter 1.0703	P3, Compressor Room	130
106	Inspected	Control Panel 3X400-V	P3, Compressor Room	131
107	Not Inspecte	Panel - Solar Panels (No Access)	P3, Extraction Room	187
108	Inspected	MDP-2 Panel 1-5	P3, LV Room	92
109	Inspected	LV Busbar (X3)	P3, LV Room	126
110	Inspected	MDP-3 (1-2)	P3, LV Room	93
111	Inspected	MDP-1 (1/1-1/6)	P3, LV Room	94
112	Inspected	Cos Phi (Comar) 90-KVAR	P3, LV Room	95
113	Inspected	Cos Phi (Comar) 210-KVAR	P3, LV Room	127
114	Inspected	Trafo 1000 KVA (Double Winding 2-TAPS) (Dry) [IR/US]	P3 (HV Room)	96
115	Inspected	Trafo 1600 KVA (Dry) [IR/US]	P3 (HV Room)	97
116	Inspected	HV Switches (Cells 1-3) (US Only) Vacuum	P3 (HV Room)	98
117	Inspected	UPS (EPA) 1	P3 (HV Room)	124
118	Inspected	UPS (EPA) 2	P3 (HV Room)	125
119	Inspected	TC Cold	P1	115
120	Inspected	Daikin Chiller	P1	120
121	Inspected	4 Motors	P1	163
122	Inspected	P1.80 Panel	P1	164
123	Inspected	TC Chaufferie HEAT-SB01	P0, Steam Generator Room	116
124	Inspected	P.0.82-1	P0, Steam Generator Room	117
125	NIS/NOP	I0700 Steam Generator	P0, Steam Generator Room	118
126	Inspected	I0782 Steam Generator	P0, Steam Generator Room	160
127	Inspected	Operating Motors	P0, Steam Generator Room	161
128	Inspected	I0704	P0, Water Treatment	119

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Equipment Summary Report

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Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
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Report ID: 390309
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Seq. No.	Status	Equipment	Location	Equip. ID
129	Inspected	Operating Motors	P0, Water Treatment	162
130	Inspected	Q106-1 ATELIER-1	Q006	60
131	Inspected	Q006	Q006	158
132	Inspected	Q014 (X2)	Q14, Workshop Store	159
133	Inspected	Q106-2 Atelier 1	Q106, Mezz	61
134	Inspected	Q007-2 (X2)	Q007 Workshop Area	156
135	Inspected	Q008-1 Stookplaats 1	Q008	67
136	Inspected	Q008-2 Distribution Panel	Q008	155
137	NIS/NOP	Bord I0035, SIHI Vacuumpompen	Q008	91
138	Inspected	Q001-1 Magazijn 1	Q001	59
139	Inspected	Workstation Cables/Plugs	Q001, Warehouse Office	154

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Recommendation Summary Report

Client: Cargill Incorporated
Division: Starches and Sweeteners
Facility: Research & Development
Location: Vilvoorde, Belgium

File No.: 495.1518
Client No.: 10779
Report ID: 390309
Date Completed: 27-May-24

Finding/ Rec No./ History	Seq. No.	Equipment	Location	Specific Item	Severity/ Impact/ Status
1 2023-04-2 **PFR**	7	Electrical, Panel Filters	General	Abnormal visual condition re: dust/dirt contamination.	FYI Low Outstanding
2 2024-05-1 New	7	Electrical, Many Cable Penetrations	General	Abnormal visual condition re: lack of fire stop packing.	FYI High Outstanding
501 2023-04-1 **PFC**	38	B430-1 HVAC Control Panel, LV_B+4-30-001-13, 39Q13	B430	Abnormal thermal condition re: the breaker.	*** Low Conf Completed

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Cargill	Vilvoorde, Belgium	WKID: 390309	File No: 495.1518

Finding No: 1 Rec No: 2023-04-2 Seq. No: 7	Temp Rise: N/A ***Severity:*** ***FYI*** Impact: Low .
Location:	General
Equipment:	Electrical, Panel Filters
Specific Item:	Abnormal visual condition re: dust/dirt contamination.









Probable Cause:	Dirty filters blocking the airflow.
Recommendation:	Filters should be cleaned or replaced frequently to avoid high temperatures in the panels.
Comments:	Both filters above are solid. Filters should be cleaned or replaced frequently. Frequency depends on rating of contamination. Not all filters have been checked but the above pictures are from the control panels at J1 and J2.

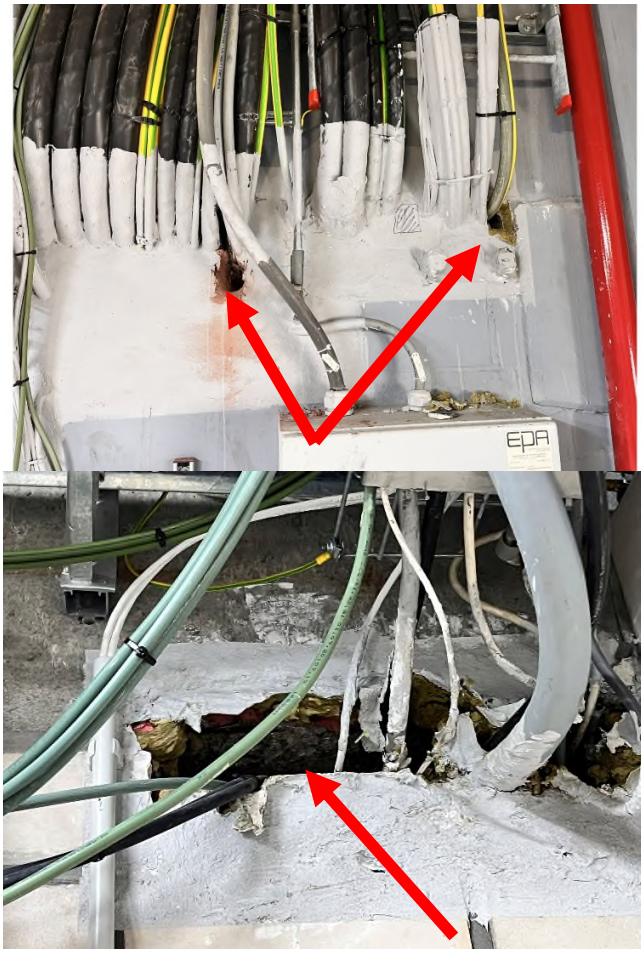
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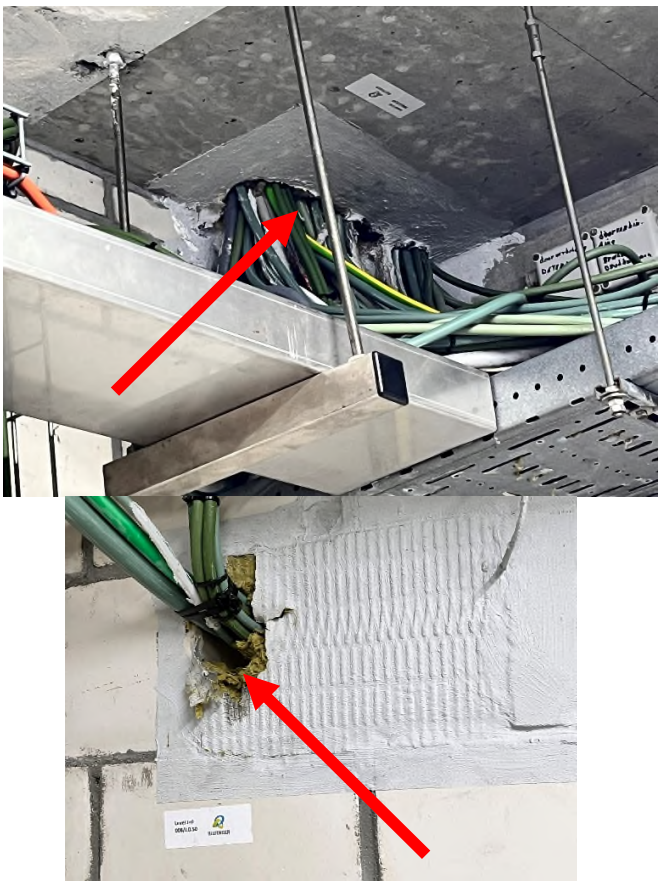
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Cargill	Vilvoorde, Belgium	WKID: 390309	File No: 495.1518

Finding No: 2 Rec No: 2024-05-1 Seq. No: 7	Temp Rise: N/A ***Severity:*** ***FYI*** Impact: High .

Location:	General
Equipment:	Electrical, Many Cable Penetrations
Specific Item:	Abnormal visual condition re: lack of fire stop packing.





Probable Cause:	Potential for collateral fire damage.
Recommendation:	Fireproof any unprotected openings with UL Listed fireproofing media having a minimum fire resistance rating of 1-hour
Comments:	Seal all openings, **make temporary fire protection when opened for projects until fire seal has been restored.** Above images are just a few examples, so not limited to just these. These images are taken at the P and J Main electrical rooms. Additional information has been provided on site.

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